

W09 · M09 · 10 MINIT / MINUTES · Hari 2, 17:10 hingga 17:20

# Hands-On SQA-AI Assistant Workshop

## Capstone with an AI pair: prove the quality gate

Kemuncak bersama rakan AI: buktikan quality gate

Let the AI evaluate a quality gate from the evidence you produced over two days, including a live test run, then present the trail and the AI mistakes you caught.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

# Before you start / Sebelum mula

## YOU WILL / ANDA AKAN

1. Run /quality-gate G3 with a live test run
2. Check every criterion has an evidence file
3. Ask the AI to list its own weakest conclusions
4. Commit your outputs as audit evidence

1. Jalankan /quality-gate G3 dengan larian ujian langsung
2. Semak setiap kriteria mempunyai fail bukti
3. Minta AI menyenaraikan kesimpulannya yang paling lemah
4. Commit outputs anda sebagai bukti audit

## CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)
- In shopfast\: npm install and npx playwright install chromium are done
- Outputs from earlier workshops exist in outputs\ (or copy them from example-output)

### Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

## OUTPUT

**outputs/m09-quality-gate.md**

W09 · STEP 1 OF 4 · LANGKAH 1

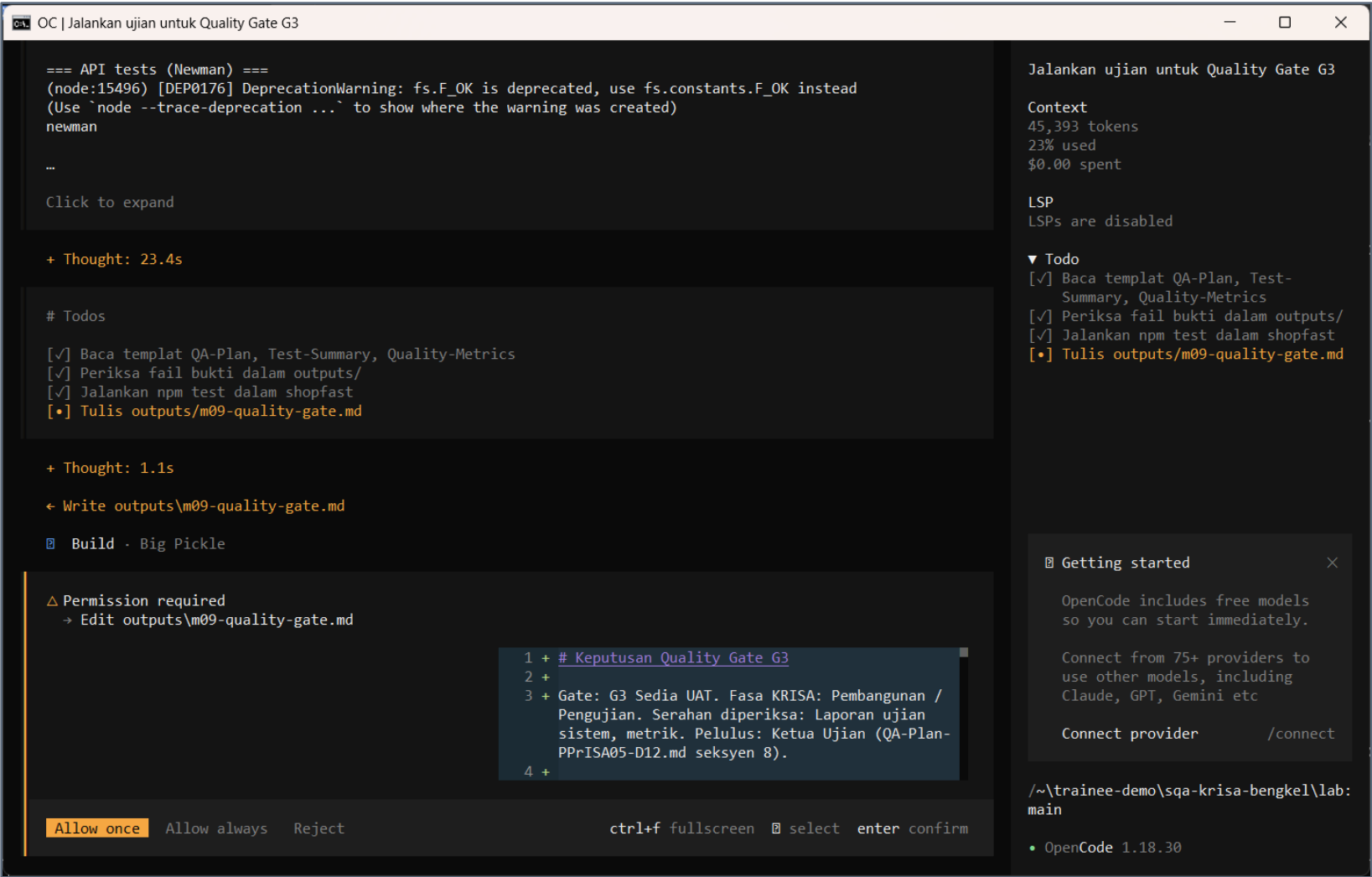
Run /quality-gate with a live test run. opencode runs npm test (allowed in opencode.json) and asks before writing. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /quality-gate dengan larian ujian langsung. opencode menjalankan npm test (dibenarkan dalam opencode.json) dan meminta kebenaran sebelum menulis. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

Salin / Copy

/quality-gate G3 jalankan-ujian



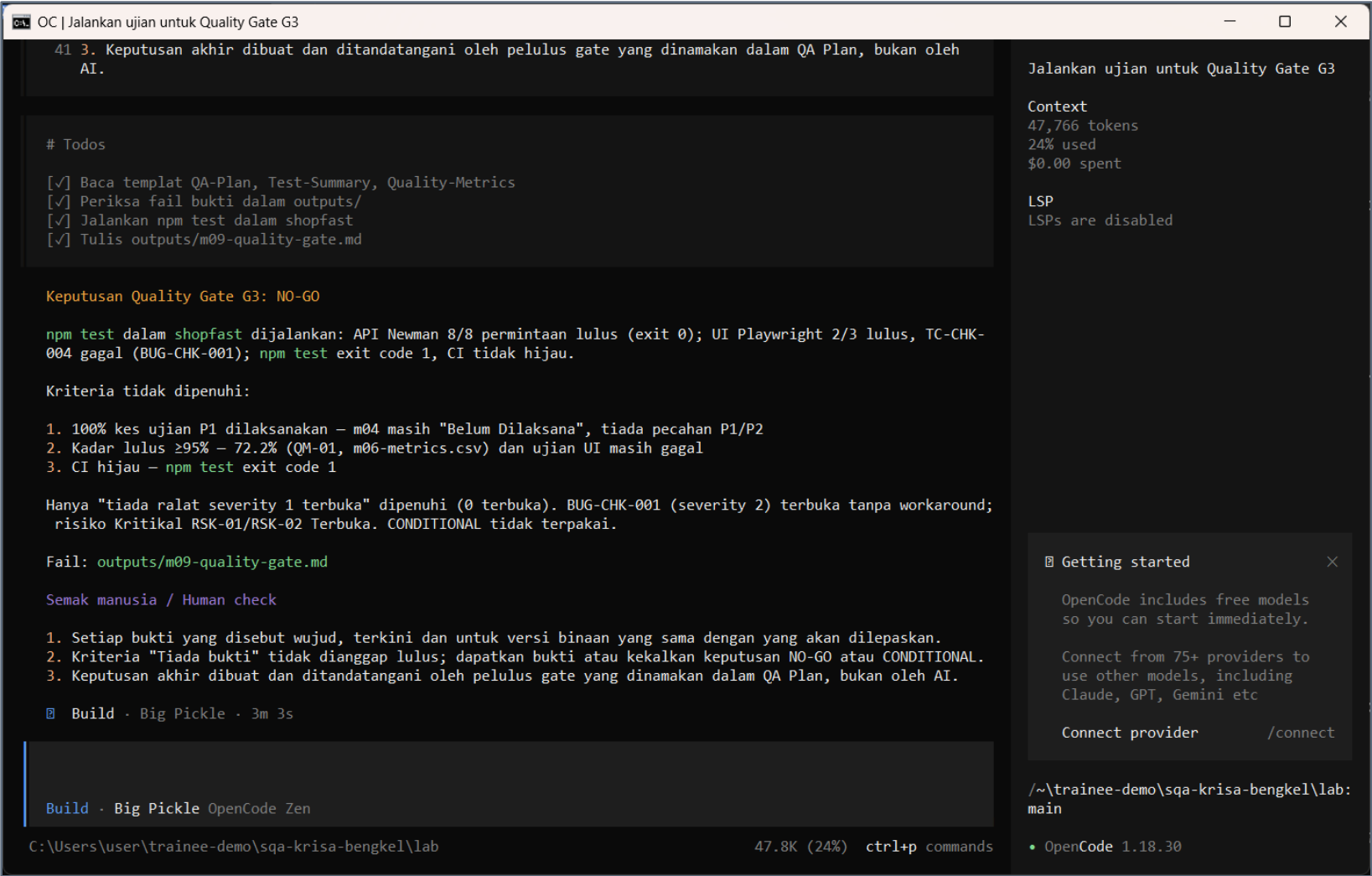
Allow once. Read the decision and the evidence per criterion.

Allow once. Baca keputusan dan bukti bagi setiap kriteria.

TYPE IN OPENCODE

Salin / Copy

Allow once (Enter)



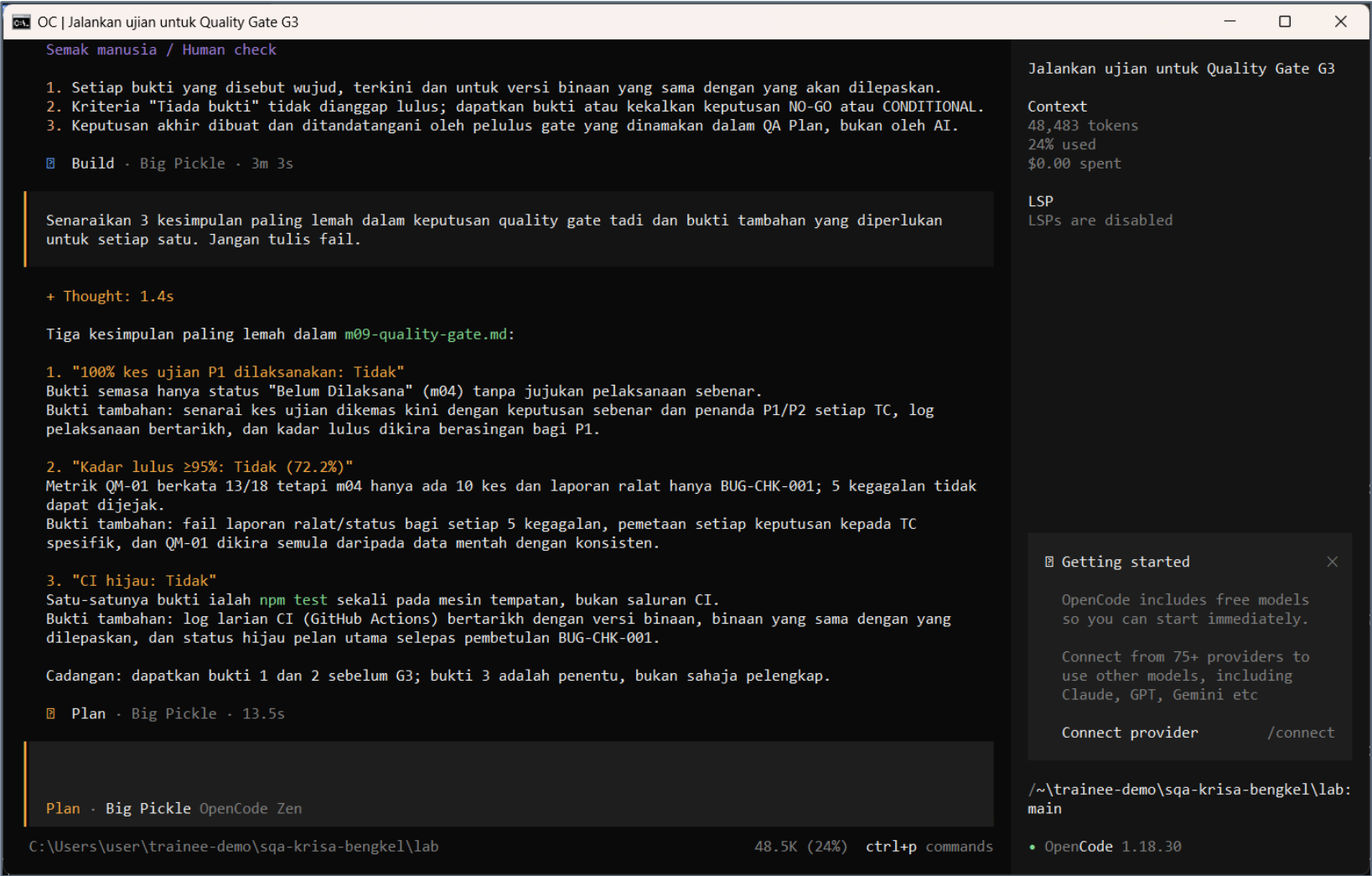
## Switch to Plan and ask the AI to criticise its own conclusions.

Tukar ke Plan dan minta AI mengkritik kesimpulannya sendiri.

TYPE IN OPENCODE (PLAN)

Salin / Copy

Senaraikan 3 kesimpulan paling lemah dalam keputusan quality gate tadi dan bukti tambahan yang diperlukan untuk setiap satu. Jangan tulis fail.



**Stage your outputs as audit evidence and commit. Use -f because lab/outputs is ignored by .gitignore, so trainee work never reaches the public repository by accident.**

Sediakan outputs anda sebagai bukti audit dan commit. Guna -f kerana lab/outputs diabaikan oleh .gitignore supaya kerja pelatih tidak termasuk ke repositori awam secara tidak sengaja.

## TYPE IN POWERSHELL

Salin / Copy

```
git add -f outputs shopfast/tests/ui/m05-
scenario.spec.js
git status --short
git commit -m "bukti kemuncak pasukan"
git log --oneline -3
```

```
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab> git add -f outputs shopfast/tests/ui/m05-senario.spec.js ; git status --short
; git commit -q -m "bukti kemuncak pasukan" ; git log --oneline -3
warning: in the working copy of 'lab/outputs/m01-glosari-qa.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m02-gwt-rewrite.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m02-rtm.csv', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m03-risk-register.csv', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m04-testcases.csv', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m05-api-tests.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m05-ui-test.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m06-bug-report-BUG-CHK-001.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m06-metrics.csv', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m07-audit.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m07-capa.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m07-test-summary.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m08-change-request.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/outputs/m09-quality-gate.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'lab/shopfast/tests/ui/m05-senario.spec.js', LF will be replaced by CRLF the next time Git touches it
A outputs/m01-glosari-qa.md
A outputs/m02-gwt-rewrite.md
A outputs/m02-rtm.csv
A outputs/m03-risk-register.csv
A outputs/m04-testcases.csv
A outputs/m05-api-tests.md
A outputs/m05-ui-test.md
A outputs/m06-bug-report-BUG-CHK-001.md
A outputs/m06-metrics.csv
A outputs/m07-audit.md
A outputs/m07-capa.md
A outputs/m07-test-summary.md
A outputs/m08-change-request.md
A outputs/m09-quality-gate.md
A shopfast/tests/ui/m05-senario.spec.js
e7fec86 (HEAD -> main) bukti kemuncak pasukan
a93ffdc (origin/main, origin/HEAD) AGENTS.md: tiada em dash atau anak panah dalam output AI
64f11e0 Bengkel AI: prompt BM Malaysia selaras arahan opencode; tolak shell ejen secara lalai; larang ekstrak PDF
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab>
```

# Human check before you accept the output / Semakan manusia

## **Every criterion links to an evidence file and a real value**

Setiap kriteria dipautkan ke fail bukti dan nilai sebenar

## **No GO while any criterion is unmet or without evidence**

Tiada GO selagi ada kriteria tidak dipenuhi atau tanpa bukti

## **At least one AI mistake was found and corrected by the team**

Sekurang-kurangnya satu kesilapan AI ditemui dan dibetulkan oleh pasukan

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

# If something goes wrong / Jika ada masalah

## **opencode is not recognised**

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

## **Scripts are disabled on this system**

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

## **The model is slow or stops**

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

## **Permission required dialog**

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

## **File or command not found**

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

## **Answer uses Indonesian words or dashes**

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

## **Network or proxy error**

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

## **Port 3000 already in use**

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`



# Abbreviations / Singkatan

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

| Abbreviation / Singkatan | Stands for / Maksud penuh                 | In plain words / Maksud mudah                                                                                                                                                                                                                     |
|--------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI                       | Artificial Intelligence                   | Software that can answer questions and write text or code by learning patterns from large amounts of data.<br><i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i> |
| API                      | Application Programming Interface         | A doorway that lets one program ask another program for data or an action, without using a screen.<br><i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>                            |
| KRISA                    | Kejuruteraan Sistem Aplikasi Sektor Awam  | The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition.<br><i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>                                          |
| MyIPO                    | Perbadanan Harta Intelek Malaysia         | Intellectual Property Corporation of Malaysia, the host of this training venue.<br><i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>                                                                                       |
| PPrISA                   | Panduan Pengurusan Projek ICT Sektor Awam | The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates.<br><i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>                                                |
| SQA                      | Software Quality Assurance                | All the planned work that gives confidence the software meets its quality requirements.<br><i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>                        |
| TUI                      | Text User Interface                       | A full-screen program that runs inside a terminal window, like opencode.<br><i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>                                                     |

# Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 8, Serahan mengikut fasa \(penyelarasan DevOps\) \(p.8\)](#)

[Bab 1, 1.6 Faktor Kejayaan \(p.14\)](#)

[PPrISA 2.0, Rajah 2.4 Pelan Tindakan \(templat mengikut fasa\) \(p.21\)](#)